

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A manufacturing plant makes **10**-inch, **9**-inch, and **7**-inch frying pans. During a certain day, the number of **10**-inch frying pans that the manufacturing plant makes is **4** times the number ***n*** of **9**-inch frying pans it makes, and the number of **7**-inch frying pans it makes is **10**. During this day, the manufacturing plant makes **100** frying pans total. Which equation represents this situation?

Answer

- A. $10(4n) + 9n + 7(10) = 100$
- B. $10n + 9n + 7n = 100$
- C. $4n + 10 = 100$
- D. $5n + 10 = 100$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day, the number of **9**-inch frying pans the manufacturing plant makes is ***n*** and the number of **7**-inch frying pans it makes is **10**. It's also given that during this day the number of **10**-inch frying pans that the manufacturing plant makes is **4** times the number of **9**-inch frying pans, or ***4n***. Therefore, the total number of **7**-inch, **9**-inch, and **10**-inch frying pans the manufacturing plant makes is ***n* + 10 + *4n***, or ***5n* + 10**. It's given that during this day the manufacturing plant makes **100** frying pans total. Thus, the equation ***5n* + 10 = 100** represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.